POWER SUPPLY PP-1588( )/MPQ-4A

1. SCOPE

1.1 This specification covers one type of power supply designated PP-1588( )/MPQ-4A, which is a unit of Radar Set AN/MPQ-4A. (See 6.1 and 6.3.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of the invitation for bids, or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Steel Strappling, Flat.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-F-320	Fiberboard: Sheet, Stock and Cut Shapes.
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

MILITARY

MIL-E-1	Electron Tubes and Crystal Rectifiers.
MIL-E-75	Electron Tubes, Packaging, Packing and Container-Marking of, General Specification for.
MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags, Interior Packing.
MIL-V-173	Varnish, Moisture and Fungus-Resistant for the Treatment of Communications, Electronic and Associated Electrical Equipment.
MIL-S-901	Shockproof Equipment, Class HI (High Impact), Shipload.
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base).
MIL-I-11748	Interference Reduction for Electrical and Electronic Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

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STANDARDS

MILITARY

MIL-STD-22	Welded-Joint Designs.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking for U. S. Military Property.
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture-Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-200	Electron Tubes.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SIGNAL CORPS

SC-A-46439	List of Accessories for Package Tester.
SC-C-85703	Test Box.
SC-DL-388060	Power Supply PP-1588( )/MPQ-4A.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Construction. — The equipment shall be constructed in accordance with Drawing SC-DL-388060 and the requirements specified herein.

3.2 Preproduction samples. — The contractor shall furnish two preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2.)

3.3 Parts, materials, and processes. —

3.3.1 Tubes, electron. — Except as otherwise provided by Standard MIL-STD-200, these electron devices shall conform to Specification MIL-E-1.

3.3.1.1 Choice, use, and application. — Except as otherwise specified in "a" and "b" below, the choice, use, and application of these electron devices shall conform to Standard MIL-STD-200:

(a) When substitution of a preferred or guidance type of electron device for a type required by the equipment specification would degrade

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equipment performance or ruggedness below specified requirements, pertinent data shall be furnished to the contracting officer, to justify retaining the type required by the equipment specification.

(b) When the foregoing standards require modification of equipment circuitry, the modification shall be subject to prior approval by the contracting officer.

3.3.1.2 Circuit location. — The electron devices furnished in equipment on contract shall be the individual devices used when the equipment was inspected for acceptance.

3.3.2 Plastic material and parts. — Where not machined, plastic material and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined shall be as smooth as practicable in accordance with good manufacturing practice for the intended application.

3.3.3 Wiring and cabling. — Wiring and cabling shall be neat and sturdy.

3.3.3.1 Slack. — Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected.

(d) To facilitate field repair of broken or cut wires.

3.3.3.2 Protection. — Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.3.3.3 Connections, general. — Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.3.3.4 Splicing. — Wires in continuous run between two terminals shall not be spliced during the wiring operation.

3.3.4 Soldering. — Soldering shall conform to the following:

3.3.4.1 Acid or acid salts. — No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of

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Project 6115-A577

electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.3.4.2 Process. — There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or under heated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.3.5 Welding. — Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary.

3.3.5.1 Cleaning prior to welding. — Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign materials.

3.3.5.2 Process. — Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum.

3.3.5.3 Cleaning after welding. — Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.3.6 Tropicalization of material. — Material shall be treated as follows:

3.3.6.1 Treating materials. — Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.3.6.2 Toxicity. — Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during the use of the finished items when used for the purpose intended.

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3.3.6.3 Flexibility. — Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.3.6.4 Statement of treatment. — The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2.)

3.3.7 Cleaning. —

3.3.7.1 Parts. — After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.3.7.2 Units. — After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.3.8 Finish, protective. — Equipment shall be given protective finish in accordance with MIL-P-14072 and the equipment drawings. (See 4.4.)

3.3.9 Marking. —

3.3.9.1 General. — Marking shall conform to Specification MIL-M-13231 and Standard MIL-STD-130. (See 4.4.) Front panel marking shall be group 1 as described in Specification MIL-M-13231.

3.3.9.2 Serial numbers. — Each power Supply PP-1588( )/MPQ-4A shall be serial numbered.

3.4 Preconditioning. — The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6. (Also see 4.5.)

3.5 Service conditions. — The equipment shall meet the following service conditions — where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.5.1 Operation. — Continuous use for a period of one year, under the worldwide environmental conditions specified in 3.5.2 through 3.5.6, with a duty cycle of four hours on and one hour off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

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3.5.2 Temperature. (See 4.7.1.)-

(a) Operating: Ambient temperature in the range of +150° F to -65°F; exposure at the high temperature extreme not to exceed four hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of +160°F to -80°F; exposure at the high temperature extreme not to exceed four hours, and at the low temperature extreme not to exceed 24 hours, at ony one time.

3.5.3 Relative humidity. — Exposure to 97 percent relative humidity for 20 hours and 100 percent relative humidity (with condensation) for four hours for each 24 hour period over the period of years stated in 3.5.1. Meeting the tests of 4.7.2 shall be considered compliance with this requirement.

3.5.4 Elevation. (See 4.7.3.) —

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

3.5.5 Orientation. (See 4.7.4.) — Operating: Any orientation up to 20 degrees from normal operating position (ie, forward, backward, left, or right).

3.5.6 Vibration, bounce and shock. —

3.5.6.1 Vibration, internal. — The amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.7.5.)

3.5.6.2 Bounce and shock. — The equipment shall be capable of meeting the requirements of table 1.

Table 1 - Bounce and shock

Requirement	Para no	Performance after test
Bounce	4.7.6	Specified performance. (Note A.) No physical damage.
Shock; ballistic	4.7.7	Specified performance. (Note A.) Any physical damage shall be minor only.
Shock; bench-handling.	4.7.8	Specified performance. (Note A.) No physical damage.

Note A: The equipment shall meet specified performance for all electrical tests. (See 4.8.1 through 4.8.4.)

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3.6 Electrical requirements. —

3.6.1 Input power. — The equipment shall operate from a 115 volt ± 10 percent 400 cycle-per-second (cps) ± 5 percent single-phase source. (See 4.8.1.)

3.6.2 Time delay circuit. — The application of the primary power to the high voltage circuits shall be delayed 30 seconds, ± 7 seconds, by a time delay circuit operated from an external 28 volt direct-current (d-c) source. (See 4.8.2.)

3.6.3 Output power. (See 4.8.3.) — With a power supply operating within the tolerances specified in 3.6.1 the equipment shall deliver output power as in 3.6.3.2 and 3.6.3.3. With an input voltage of 115 volts ac the ac outputs shall be as in 3.6.3.1.

3.6.3.1 Alternating-current power. — The equipment shall have the following a-c outputs at 115 volt a-c input:

(a) 6.4 volts ac ±1 percent at 2.6 amperes (amp).
(b) 6.4 volts ac ±1 percent at 2.6 amp.
(c) 6.4 volts ac ±1 percent at 2.6 amp.

3.6.3.2 Regulated d-c power. — The equipment shall have the following d-c outputs:

(a) Nominal +300 volts at 85 milliampere (ma) and within ±10 volts from 25 percent underload to 25 percent overload.
(b) Nominal +150 volts at 190 ma and within ±5 volts from 25 percent underload to 25 percent overload.
(c) Nominal -300 volts at 30 ma and within ± 10 volts from 25 percent underload to 25 percent overload.

3.6.3.3 Keep-alive voltage. — The equipment shall have a keep-alive voltage to the TR tube of +620 ±20 volts at 100 microamperes.

3.6.3.4 Root mean square (rms) ripple voltage. — The rms ripple voltage on each d-c output under normal load and a 115-volt ac line voltage shall not exceed 15 millivolts (mv.) (See 4.8.3.1.)

3.6.4 Overload protection. — During normal operation removal of the overload protection fuses shall cause the appropriate blown-fuse indicators to light. (See 4.8.4.)

3.6.5 Radio Interference suppression. — The equipment shall conform to Specification MIL-I-11748 and the class III limits specified therein. (See 4.8.5.)

3.7 Interchangeability. — Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.9.) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

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3.8 Final operation. — Just prior to packaging, the equipment shall be examined visually and electrically. (See 4.10.)

3.9 Technical literature, tools, and running spare parts. — Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.10 Workmanship. — The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.3.1 Electron Tubes.
3.3.2 General requirements for plastic material and plastic parts.
3.3.3 Wiring and cabling, including:
- 3.3.3.1 Slack.
- 3.3.3.2 Protection.
- 3.3.3.3 Connections, general.
- 3.3.3.4 Splicing.

3.3.4 Process for soldering.
3.3.5.1 Cleaning of welding.
3.3.5.2 Process for welding.
3.3.7 Cleaning

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. — Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. The Government reserves the right to perform only of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. — Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery), (See 4.3.)

(b) Acceptance inspection.

(1) Acceptance inspection of equipment before preparation for delivery. (See 4.4 and 4.5.)

(2) Acceptance inspection of preparation for delivery. (See 4.12.)

4.3 Preproduction inspection. — This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table II, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A and group B (see tables III and IV, respectively). The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, (3) shock, bench-handling and, (4) shock, ballistic; other preproduction inspection may precede, following, or be interspersed between the foregoing.

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Table II - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3.)	Reg para	Insp para
Temperature	3.5.2	4.7.1
Moisture-resistance	3.5.3	4.7.2
Elevation	3.5.4	4.7.3
Orientation	3.5.5	4.7.4
Vibration	3.5.6.1	4.7.5
Bounce	3.5.6.2	4.7.6
Shock, ballistic	3.5.6.2	4.7.7
Suppression of radio interference	3.6.5	4.8.5

4.4 Acceptance inspection covered by subsidiary documents. — The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Finish	3.3.8
Marking	3.3.1

4.5 Acceptance inspection of equipment before preparation for delivery. — The contractor, to demonstrate compliance with specified requirements, shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.4.) Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A and group B inspection shall be preconditioned after final assembly. (See 3.4.)

4.5.1 Group A inspection. — This inspection, including sampling, shall conform to table III and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the final operation inspection shall be last. (See 4.10.)

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Table III - Group A inspection

Inspection	Reg para	Insp para	AQL	
			Major	Minor
Visual and mechanical	3.10	4.11	1.0%	4.0%
Electrical				
Input power	3.6.1	4.8.1	4 dphu* for the group	
Time delay circuit	3.6.2	4.8.2		
Output power	3.6.3	4.8.3		
Ripple voltage	3.6.3.4	4.8.3.1		**
Overload protection	3.6.4	4.8.4		
Final operation	3.8	4.10	1.0%	

* Defects per hundred units.
**All electrical defects and final operation are considered major.

4.5.2 Group B inspection. — This inspection, including sampling, shall conform to table IV and to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

Table IV - Group B inspection

Inspection	Reg para	Insp para
Shock, benchhandling	3.5.6.2	4.7.8
Interchangeability	3.7	4.9

4.5.3 Reinspection of conforming group B sample units. — Unless otherwise specified, sample units which have been subjected to and passed group B inspection, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

4.6 Bounce preconditioning. — The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L. A. B. Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of three inches as the unit bounces. The unit shall be subjected to this preconditioning for one minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

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4.7 Service conditions tests. —

4.7.1 Temperature tests. (See 3.5.2.) — The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Performance shall be as follows:

Steps	Paragraph	Measurement	Performance
1, 3, 7 and 10	4.8.1 through 4.8.4	Electrical tests	Full specification requirements

4.7.2 Moisture-resistance tests. (See 3.5.3.) — The equipment shall be tested as follows:

4.7.2.1 Test conditions. —

(a) Do not remove equipment from the humidity chamber for measurements.

(b) Start measurements not more than five minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.7.2.2 Test method. —

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.7.2.3 and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.7.2.3.

4.7.2.3 Performance. — Measurements made during test and specified performance during cycling shall be as follows:

Paragraph	Measurement	Performance during cycling
4.8.1 through 4.8.4	Electrical tests	Full specification requirements

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4.7.2.4 Failure. — If any equipment fails to meet the performance specified in 4.7.2.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.7.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.7.2.2 (e), it does not pass the test.

4.7.3 Elevation test. (See 3.5.4.) — The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	Measurement
4.8.1 through 4.8.4	Electrical tests

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperatures and pressure stabilization, repeat the measurements specified in 4.7.3(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 ± 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for one hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment, and after stabilization, repeat the measurements specified in 4.7.3(a). The equipment shall meet specified performance for these measurements.

4.7.4 Orientation test. (See 3.5.5.) — The equipment shall be inclined for a minimun of five minutes in each plane (forward, backward, left, and right) to an angle of 20 (+3 -1) degrees. During inclination in each plane, the equipment shall meet specified performance for the following measurements.

Paragraph	Measurement
4.8.1 through 4.8.4	Electrical tests

4.7.5 Test for internal vibration. — Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.5.6.1:

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate the observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

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(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.7.6 Bounce test. (See 3.5.6.2.) — The equipment shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

(a) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at six-inch intervals around all four edges and at three-inch intervals in a six-inch square in the center.

(b) Place the equipment in its transit case prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than three inches and not less than one inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of three hours at 284 ± 2 rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.7.7 Shock test; ballistic. (See 3.5.6.2.) — The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The equipment shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of nine blows: one each one-foot blow, three-foot blow, and five-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible.

4.7.8 Shock test; bench-handling. (See 3.5.6.2.) — The chassis-and-front-panel assembly shall be removed from its enclosure, as applicable, and placed in a suitable position for servicing on a solid, two-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

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(b) Repeat "(a)", with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembl might practicably be placed during servicing.

4.8 Electrical tests. —

4.8.1 Test setup. — The equipment shall be connected to a dummy load, constructe in accordance with Drawing SC-C-85703. The dummy load shall be connected to an adequate 115-volt, 400-cps source and a 28 volt d-c source. The alternating current source shall be capable of being varied from 103.5 to 126.5 volts and from 380 to 420 cps. (See 3.6.1.) Unless otherwise stated herein the bias control of the regulator circuit shall be set to supply -300 volts out with a 115 ± 2 volt a-c supply and a normal load across the output terminals.

4.8.2 Time delay. — Apply 115-volt, 400-cycle power. Apply the 28-volt d-c powe (the a-c primary voltage shall be applied simultaneously with or prior to the d-c voltage), and with a stop watch measure the time it takes for output dc voltages (indicated by a continuity checker) to appear at the output terminals. The results shall conform to 3.6.2.

4.8.3 Output voltage test. — The output voltages shall be measured with the dummy load set for normal load and the input frequency varied gradually from 380 to 420 cps at the following input a-c supply voltages:

103.5 volts, 115 volts, 120 volts, and 126.5 volts.

The a-c outputs shall be measured at the 115 volt input only.

Repeat the measurements with 25 percent overload and 25 percent underload. The keep-alive voltage terminals shall be connected to a resistive load of 100 microamperes as measured by a microammeter with an accuracy within ±0.3 percent of full scale placed between the terminal and a resistive load. The voltage developed across the load shall be measured using a DC differential voltmeter which has an accuracy within ±0.3 percent and an input impedance not less than 10 megohms. The results shall conform to 3.6.3.

4.8.3.1 Ripple voltage test. — The input power shall be set at 115 volts and the dummy load set for normal load. The ripple voltage shall be measured with a multimeter having an accuracy of 2 percent throughout voltage and frequency ranges and at all points on the meter scale. The ripple voltage present at each d-c output terminal, shall conform to 3.6.3.4.

4.8.4 Overload protection test. — With the equipment operating and the dummy load adjusted for normal load, remove the fuses from their respective circuits. Observe the blown fuse indicators for conformance to 3.6.4.

4.8.5 Radio interference suppression. — The equipment shall be tested for unwanted rf radiation in accordance with the requirements of 3.6.5.

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4.9 Inspection for interchangeability. — The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.7. Where a listed dimension is not within specified or design limits it shall be considered a major defect.

(a) External dimensions of insertable assemblies and internal dimensions of cabinet.

(b) Location of hinges and fasteners on cabinet and fasterners, slides, and mountings of assemblies.

4.10 Final operation test. — Prior to packaging the equipment shall be checked visually for damage, loose parts and completeness. All electrical circuits shall be checked for apparent malfunction. (See 3.8.)

4.11 Visual and mechanical inspection. — The units shall be examined for the defects listed in Standard MIL-STD-252.

4.12 Acceptance inspection for preparation for delivery. — Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

4.13 Rough handling test (preparation for delivery). — When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2), the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

4.8.1 through 4.8.4 — Electrical tests

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. —

5.1.1 Level A. — Each Power Supply PP-1588( )/MPQ-4A shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116, and described in the following paragraphs.

5.1.1.1 Technical manual. — Each technical manual shall be packaged Method IC-6 as follows: Inclose each technical manual within a close-fitting Type II, Class d bag, conforming to Specification MIL-B-117. Heat seal to effect closure.

5.1.1.2 Tubes. — All tubes over four inches in length, not securely clamped in place, shall be removed, and together with spare tubes packaged individually in accordance with the applicable provisions of Specification MIL-E-75.

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5.1.1.3 Accessories and spare parts items. — Accessories and spare parts items, determined in quantity and number from the bid request or contract, shall be packaged in accordance with the applicable provisions of Specification MIL-P-116.

5.1.1.4 Consolidated package. — Consolidate the items packaged as specified in 5.1.1.2 and 5.1.1.3 together within a close-fitting, Style RSC, Class 2, W5c fiberboard box, conforming to Specification PPP-B-636. Closure shall be in accordance with the appendix of the box specification. In addition, all seams and joints shall be sealed with water resistant, pressure sensitive tape, conforming to Specification PPP-T-76.

5.1.1.5 Power Supply PP-1588( )/MPQ-4A. — Each power supply shall be packaged Method IC-5 as follows: Cushion each power supply on all surfaces with cells or pads or both fabricated of fiberboard, Type CF, Class weather resistant, Variety SW, Grade W5c, conforming to Specification PPP-F-320. Place the cushioned power supply within a close-fitting, Style RSC, Class 2, W5c fiberboard box, conforming to Specification PPP-B-636. Closure shall be made in accordance with the appendix of the box specification. In addition, all seams and joints shall be sealed with water resistant, pressure sensitive tape, conforming to Specification PPP-T-76.

5.1.2 Level C. — Each Power Supply PP-1588( )/MPQ-4A shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.1.3 Package performance. — Package testing shall be performed in accordance with the requirements of Specification MIL-P-116. The rough handling and functional tes. will be required only when invoked in the bid request or contract. (See 6.2.)

5.2 Packing. —

5.2.1 Level A. — Each Power Supply PP-1588( )/MPQ-4A shall be packed as follows: Place the consolidated package specified in 5.1.1.4 and the power supply specified in 5.1.1.5 together within a Class 2, Style 4, nailed wood box or Class 2, Type III, Style A, cleated plywood box conforming to Specification PPP-B-621 or PPP-B-601, respectively. Box contents shall be blocked and braced to prevent movement inside the container. Fabricate the box to fit contents snugly. Place the technical manual, packaged as specified in 5.1.1.1, on top of the contents directly under lid of the box. Closure shall be in accordance with the appendix of the applicable box specification.

5.2.1.1 Metal strapping. — Metal strapping conforming to Specification QQ-S-781, Type 1, Class b, shall be applied to the wooden shipping container in accordance with the requirements of the applicable box specification and appendix thereof.

5.2.2 Level B. — Each Power Supply PP-1588( )/MPQ-4A shall be packed as specified in 5.2.1, except the nailed wood box shall be Class 1 and the cleated plywood box shall be Type IV, Class 1.

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5.2.3 Level C. — Each Power Supply PP-1588( )/MPQ-4A shall be packed in shipping containers of the type, size and kind commonly used for the purpose, in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier applicable to the mode of transportation.

5.3 Marking. — Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Intended use. — Power Supply PP-1588( )/MPQ-4A furnishes d-c operating voltages to the receiving system and the trigger amplifier, and furnishes keep-alive voltages to the TR tube of Radar Set AN/MPQ-4A. It may also be used as part of Test Facilities Kit MK-387/MPQ-4A.

6.2 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Levels of preservation, packaging and packing required for shipment, (Level A, level B or level C.)

(c) The specific paragraphs of section 5 which are applicable to the particular procurement.

(d) Preproduction inspection:

Preproduction pack(s) as follows:

Makeup of pack(s).

Number of each kind of pack to be submitted.

Inspection to be performed thereon.

(e) Marking and shipping of samples.

(f) Place of final inspection.

(g) Technical literature required. (See 3.9.)

(h) Quantity of tools and running spare parts required. (See 3.9.)

(i) Submission of the statement of treatment referenced in 3.3.6, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

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(j) If rough handling and functional tests are required. (See 4.13.)

6.3 Nomenclature. — The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-1588A/MPQ-4. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.4 Verification inspection. — Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product, and will normally be identified by the categories listed below:

(a) Type A — The total of that inspection set forth in the Quality Assurance Provisions of this specification or the contract. Included in this category is that amount of inspection referred to as normal and tightened inspection by Military Standard 105.

(b) Type B — That inspection set forth in the Quality Assurance Provisions of this specification or the contract reduced in amount under the reduced inspection provisions of Military Standard 105.

(c) Type C — A reduced inspection procedure resulting in a material reduction in the amount of inspection set forth in the Quality Assurance Provisions of this specification. The amount of inspection is less than that provided for in type B and is based upon a consistently acceptable product resulting from a planned quality control system voluntarily employed by the contractor in the production of the product.

NOTICE: When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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